Domain-aware heterogeneous quantization and causal sub-network probes

`pretty_duck`, branch `feature/domain-compression`

Abstract

We describe a pipeline that (i) collects per-domain activation statistics of a language model, (ii) plans a heterogeneous quantization of its linear layers under a byte budget, and (iii) writes the plan as real packed weights with a measured manifest. On top of the statistics we add causal probes (ablation and minimal-pair activation patching) that ask whether channels selected by a domain contrast are *used* by the model, rather than merely more active on that domain. The results so far are mixed and are reported as measured. At block and row-group granularity the domain signal is too weak to change an allocation. Per input channel it is reproducible, but a mean-ablation effect that looks selective is fully explained by the size of the perturbation. On Persona-style prompt pairs, however, a mean-difference channel selector finds channels that are both sufficient and necessary for the prompt-induced margin shift, beyond random and beyond a lexical control. Quantizing those channels at 3–4 bits does not damage that behavior more than quantizing random channels of the same cost.

1 Overview

The pipeline has three stages (Figure 1). This document covers the first two and the converter's storage format; the final inference pipeline is out of scope. No gradients, auxiliary networks or fine-tuning are used anywhere, and every probe is forward-only.

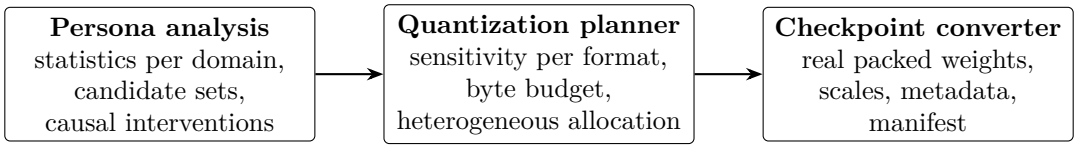


Figure 1: The three stages. The causal probes of Section 4 belong to the first stage; only candidate sets that survive them would be passed to the planner.

2 Statistics and local quantities

Let g index the `nn.Linear` layers of the text tower, $W_g \in \mathbb{R}^{o\times i}$ the weight and $h_g(x) \in \mathbb{R}^i$ its input. For a domain d with calibration set C_d we accumulate, over valid (non-padding) tokens and in double precision, three per-input-channel moments:

$$A_{g,d,j} = \mathbb{E}\,|h_{g,j}|, \qquad V_{g,d,j} = \mathbb{E}\,h_{g,j}^2, \qquad M_{g,d,j} = \mathbb{E}\,h_{g,j}. \tag{1}$$

Hooks are removed on exit, also after an exception; tied heads and non-text towers are skipped. The per-block importance and the diagonal quantization-error estimate are

$$I_{g,d} = \sum_{i,j} |W_{g,ij}|\, A_{g,d,j}, \tag{2}$$

$$D_{g,d}(b) = \sum_{i,j} \big(W_{g,ij} - Q_b(W_{g,ij})\big)^2\, V_{g,d,j} \;\approx\; \mathbb{E}_{x\sim C_d}\big\|x\,E_g^{(b)\top}\big\|_2^2, \tag{3}$$

where the last step assumes uncorrelated input channels. D is a local prioritization signal, not a prediction of the final loss: errors in different layers are not equally harmful, and the sum over layers is not the loss increase.

3 Quantization planner

Formats and real bytes. Four portable formats are supported, all quantizing each output row independently: `fp16`; `int8` and `int4` (symmetric, one fp16 scale per row); and `nf4` (NF4 codebook, one fp16 absmax per 64-weight block of a row). The allocator uses exact payload formulas, asserted against the packed tensors, never nominal bits per parameter: for a (o, i) weight, $B(\texttt{int8}) = oi + 2o$, $B(\texttt{int4}) = \lceil oi/2 \rceil + 2o$ and $B(\texttt{nf4}) = o\lceil i/64 \rceil\,(32 + 2)$.

Greedy allocation. Every unit starts at `fp16`. A reduction $b \to b'$ of a unit g that saves bytes has cost

$$\rho_{g,b\to b'} = \frac{D_{g,T}(b') - D_{g,T}(b)}{B_g(b) - B_g(b')}, \tag{4}$$

and the smallest ρ is applied until a size budget is met, or until no reduction fits an error budget. Steps that save no bytes or give a non-finite ρ are skipped. The plan is a reproducible JSON file, separate from its application to a model. Greedy selection is optimal only on the lower convex hull of each block's (bytes, error) points, which is not guaranteed.

Row-group units. A unit is a whole layer or a group of consecutive output rows, named `layer@r0:r1`. Because each format treats rows independently, a group quantized on its own is bit-identical to the same rows inside the full matrix, and its bytes are exactly $B((r, i), b)$; this is checked by tests, including odd widths where int4 rows share a byte. The converter regroups units into one file per layer and rejects groups that do not tile the layer.

Contrast signals. Two auxiliary signals can multiply ρ. The first uses $C_g = I_{g,T} - \lambda\,\overline{I_{g,S}}$, min–max normalized over blocks. The second, *target-anchored relative contrast* (TARC), ranks each domain across units, $R_d(g) = \mathrm{ranknorm}(I_d(g)) \in [0, 1]$ (average ranks for ties), and takes

$$S_T(g) = R_T(g) - \mathrm{median}_{d\neq T}\, R_d(g) \in [-1, 1]. \tag{5}$$

Block size and layer type move every domain's rank the same way and cancel in the difference; the penalty is $\rho \leftarrow \rho\,(1 + \lambda \max(S_T, 0))$, so λ is only a strength. Both are hypotheses about prioritization: the error term already contains the domain through $V_{g,T}$, and only evaluation can say whether the contrast adds anything.

4 Causal probes

Statistics say a channel is more active on a domain; they do not say the model needs it. The probes intervene.

4.1 Candidate channel sets

For each layer a fraction k of its i input channels is selected by one of:

- **tarc**: top channels by S_T computed from the per-channel $A_{d,j}$ ranks within the layer;
- **zdiff**: top channels by $|M_{T,j} - \overline{M}_{o,j}|/\sqrt{\mathrm{var}_{T,j} + \overline{\mathrm{var}}_{o,j}}$, with $\mathrm{var} = V - M^2$ and bars denoting the mean over the other domains (the standardized mean difference);
- **importance**: top channels by $A_{T,j}$ alone; **anti**: bottom channels by S_T;

- **random**: same size, disjoint from a reference set, matched channel by channel on the V-rank;
- **energy-random**: same size, disjoint, matched on the pair $(\log e_j^T, \log e_j^o)$ defined next.

4.2 Mean ablation and the dose confound

Replacing channel j by a value m_j (the mean of the other domains' M) perturbs the input by an expected squared amount $e_j = \mathbb{E}(h_j - m_j)^2 = V_j - 2m_jM_j + m_j^2$ on each domain. A set chosen because the target differs from the others perturbs the target text more *by construction*. With Δ_d the increase in mean NLL on held-out text of domain d, we report the selectivity $\Delta_T - \overline{\Delta}_o$ with a bootstrap interval over lines, and compare it with random and energy-matched sets. Equal-dose controls are essential: a selective effect that they reproduce is not evidence of necessity.

4.3 Minimal-pair activation patching

A pair is one question and answer option scored under two system prompts, *clean* and *corrupt* (for example the seek and reject personas). The margin of a run is

$$m = \log p(\text{persona-matching option}) - \log p(\text{other option}), \tag{6}$$

with log-probabilities summed over the answer tokens. The question and answer tokens are identical, so the two token sequences share a common suffix and positions are aligned from the end. We run the clean sequence, record every `nn.Linear` input over that suffix, and rerun the corrupt sequence while overwriting the selected channels with the clean values. The *restoration* of a set is

$$\text{rest} = \frac{\sum_q (m_q^{\text{patched}} - m_q^{\text{corrupt}})}{\sum_q (m_q^{\text{clean}} - m_q^{\text{corrupt}})}, \tag{7}$$

with a percentile bootstrap over questions. Swapping the roles of the prompts gives the *necessity* direction: the fraction of the shift destroyed when the other prompt's activations are patched in.

Built-in reference. Patching *every* Linear input over the suffix reproduces the clean run exactly: each residual contribution at those positions then comes from a clean input, so the residual stream and the logits there equal the clean run's. This holds when no parameterized non-Linear path writes to the residual stream, and was verified numerically (restoration 1.000).

Controls. Besides random sets, every set is evaluated on three prompt pairs scored on the same questions: the *real* pair (seek/reject), a *shuffled* pair (the same words in random order, which keeps lexical content and destroys the instruction) and an *irrelevant* pair (two unrelated prompts of similar length). A set is persona-related only if it restores the real shift more than the shuffled one, with a paired interval, and not the irrelevant one.

4.4 Quantization as an intervention

Input channel j is column j of W. For a channel set we replace those columns by their quantize–dequantize round trip, with per-row scales computed over the selected columns only, and leave everything else at full precision. Each condition reports its diagonal dose $\sum_{i,j\in\text{set}} E_{ij}^2 V_{T,j}$ so that sets can be compared at equal cost, and the *retention* $\sum_q (m_q^{\text{seek}} - m_q^{\text{reject}}) / \sum_q (m_q^{\text{seek},0} - m_q^{\text{reject},0})$ of the unquantized shift. Bit widths below 4 are stronger interventions only; nothing is packed or stored.

5 Experimental setup

- **Models.** SmolLM-135M (CPU, bf16) for statistics, block and row-group analyses and mean ablation; Qwen3.5-0.8B (RTX 3090, fp16) for the persona experiments.
- **Calibration and evaluation text.** MMLU *validation* questions with choices (no answer keys) for calibration; held-out MMLU *test* text for evaluation. Domains: mathematics (92 samples), law (194), biology (80), computer science (42), humanities (97), each also split into two disjoint halves. Persona: the 623 rows of `wealth_train.csv` give seek (matching) and reject (not-matching) texts; evaluation uses the 200 questions of `wealth_test_infer.csv`, scored with the original wealth prompts.
- **Seeds and sizes.** Seed 42 for sample selection; 5 draws for each random control; bootstrap intervals use 500–2000 resamples; mean ablation uses 30 held-out lines per domain.

6 Results

6.1 Block-level and row-group signals (SmolLM-135M)

With math and law over the 210 blocks, the rank correlation between the two domains' errors is 0.995 (int8, int4) and 0.996 (nf4), and between importances 0.996. The per-block ratio $D_{\text{math}}/D_{\text{law}}$ (nf4) ranges from 0.49 to 2.8 with median 1.02. These raw sums are dominated by block size, which motivated TARC.

With five domains split in halves, math-anchored S_T has standard deviation 0.023 over blocks and range $[-0.14, 0.13]$ (split-half Spearman 0.91); at 64-row groups (2430 units) it is 0.030, range $[-0.25, 0.19]$ (split-half 0.93). It is reproducible but small. Table 1 compares plans at a size budget 40% of the way from all-int4 to fp16 (about 117 MB of 212 MB). Plans barely

	whole layers (210)		64-row groups (2430)	
TARC λ	units changed	est. target error	units changed	est. target error
1	1	−1.84%	0	+0.00%
10	1	−1.84%	13	+0.78%
100	7	+10.60%	33	+2.06%
1000	39	+87.37%	277	+60.03%
math vs law target (error-only)	2	–	22	–

Table 1: Plans against the error-only plan at equal size; "est. target error" is the sum of D_T relative to it. The negative entry shows the plain greedy is not optimal.

depend on the target domain, and the penalty only moves units at strengths that raise the estimated error.

Per input channel the signal is wider: S_T has standard deviation 0.198, 20.5% of channels have $|S_T| > 0.25$ and the split-half correlation is 0.84. The ratio $V_{\text{math}}/\operatorname{median} V_{\text{others}}$ has 5/25/50/75/95th percentiles 0.58/0.88/1.04/1.25/2.17; 13.1% of channels exceed 1.5 and carry 16.3% of math's second-moment mass.

6.2 Mean ablation

Table 2 shows selectivity at $k = 1\%$ and 2% of channels per layer. The top-A_T (importance) set is catastrophic and not selective (about 6 nats on every domain; the set holds roughly half of the V mass), and the *anti* set is the mirror image of tarc. Both are consistent with a dose effect. The gap tarc − energy-random is +0.009 and +0.012 (math) and +0.008 and −0.006 (law), inside the spread of the controls. The earlier selectivity is explained by dose.

anchor	k	tarc	random (V-matched)	energy-random
math	1%	+0.100	+0.007 [−0.008, +0.027]	+0.091 [+0.081, +0.105]
math	2%	+0.156	+0.022 [−0.054, +0.107]	+0.144 [+0.126, +0.166]
law	1%	+0.087	+0.005 [−0.008, +0.020]	+0.079 [+0.059, +0.101]
law	2%	+0.135	−0.061 [−0.285, +0.046]	+0.141 [+0.118, +0.156]

Table 2: Selectivity $\Delta_T - \overline{\Delta}_o$ in nats (mean of 5 draws, range in brackets). tarc beats the V-matched control but not the energy-matched one.

6.3 Does a persona prompt move the margin?

On the 200 wealth questions the seek-minus-reject shift of the margin is clearly positive in both models (Table 3), but the fraction of questions where the model prefers the matching answer stays near chance under every prompt (0.52–0.55), so accuracy hides the effect and only the paired shift is used. The absolute margin is biased by answer length, which the paired difference cancels. For SmolLM the shuffled prompt shifts the margin as much as the real one, so it

model	real (seek−reject)	shuffled words	irrelevant pair
SmolLM-135M	+0.83 [+0.53, +1.11]	+0.97 [+0.63, +1.29]	+0.23 [+0.00, +0.45]
Qwen3.5-0.8B	+2.82 [+2.31, +3.34]	+1.95 [+1.53, +2.36]	+0.91 [+0.51, +1.31]

Table 3: Shift of the margin $\log p(\text{matching}) - \log p(\text{other})$ in nats with 95% intervals.

responds only to word overlap and is not used further. For Qwen the persona-specific part (real minus shuffled) is +0.85 [+0.45, +1.28] nats, a minority of the shift.

6.4 Minimal-pair patching (Qwen3.5-0.8B, $k = 5\%$, 13,782 channels)

The ceiling (all Linear inputs) restores 1.000 for the real and shuffled pairs and 0.999 for the irrelevant pair. *zdiff* is both sufficient (0.84 vs 0.18 for random) and necessary (0.80 vs 0.15),

	sufficiency (restoration)			
set	real	shuffled	irrelevant	paired real−shuffled, nats
zdiff	0.84	0.63	0.23	+1.11 [+0.85, +1.37]
tarc	0.64	0.43	0.75	+0.94 [+0.66, +1.21]
importance	0.15	0.16	0.51	+0.09 [−0.18, +0.38]
random (5)	0.18	0.18	0.17	+0.15 [−0.01, +0.29]

Table 4: Fraction of the clean-minus-corrupt margin shift restored by patching each set; the last column is the paired difference in nats restored on the same questions. Random ranges: 0.08–0.29 (real).

and its effect on the unrelated pair is at random level; its paired real-minus-shuffled difference (+1.11 nats) is clearly nonzero and at least as large as the whole persona-specific shift (+0.85). *tarc* restores the unrelated pair as much as the real one, so it picks generically context-sensitive channels, and *importance* is dominated by high-activation channels. Patching a single layer's inputs, layers 11–19 restore 13–43% of the real shift (layer 12 best, 0.43), layers 0–10 and 20–22 restore about nothing, and layer 23 is negative (−0.53). At $k = 20\%$ every set saturates (random restores about 0.45).

set	real	shuffled	irrelevant
zdiff	0.80 [0.72, 0.89]	0.67 [0.57, 0.80]	0.28 [0.05, 0.50]
tarc	0.59 [0.50, 0.69]	0.56 [0.44, 0.69]	0.72 [0.52, 1.07]
importance	−0.02 [−0.17, 0.10]	0.16 [0.01, 0.27]	0.36 [0.04, 0.60]
random (5)	0.15	0.18	0.15

Table 5: Necessity: fraction of the shift destroyed when the other prompt's activations are patched into the run, with 95% intervals.

6.5 Quantizing the selected columns

Table 6 gives the retention of the seek-minus-reject shift (2.82 nats unquantized) when only the selected columns are fake-quantized. Doses of the zdiff and random sets are within about 10% of each other (for example 75 vs 82 at int4, $k = 5\%$). At 3 and 4 bits there is no effect: the

bits	k	zdiff	tarc	importance	random (5)
int4	5%	1.01	0.96	0.89	1.00
int4	20%	0.99	1.06	0.68	1.02
int3	5%	1.08	1.04	0.87	0.96
int3	20%	1.09	1.12	0.34	1.01
int2	5%	0.73	0.78	−0.47	0.89
int2	20%	0.21	0.46	0.56	0.41

Table 6: Retention of the persona margin shift. Quantizing all columns gives 0.82 [0.65, 1.05] at int4 and about 0 at int3 and int2. Intervals around single entries are roughly ±0.1–0.3, and the random draws at int2, $k = 20\%$ range from 0.08 to 0.65.

persona shift survives quantizing the persona columns entirely. At 2 bits *zdiff* loses somewhat more than random in both settings, but the random draws vary too much to call it a result; the importance set loses a lot, at ten times the dose.

7 Discussion and limitations

What the evidence supports. (i) Domain-contrast signals are real but small at block and row-group granularity and do not change a size-matched allocation. (ii) Mean ablation with a domain-contrast set looks selective, but an equal-dose control reproduces it. (iii) On minimal pairs, the standardized-mean-difference set (*zdiff*) is sufficient and necessary for the prompt-induced margin shift, specific to the persona content over an unrelated pair, and partly beyond word overlap. (iv) That does not translate into a quantization signal at 3–4 bits.

What it does not show.

- One model per experiment, one trait (wealth), one seed; channels are chosen on the training split and tested on held-out questions.
- The effect is a log-probability shift; the model's choices stay near chance, and about two thirds of the shift is word-level priming.
- Patching covers the question and answer tokens, including the scored tokens, so it shows where an effect is expressed, not where a persona is stored.
- Only the diagonal approximation of quantization error is used; bit widths below 4 are simulations; per-row scales in the column intervention use only the selected columns.

- Factual knowledge was not tested. A knowledge claim would need a model that answers above chance, a knowledge metric (logit difference of correct and incorrect answers), minimal pairs with an entity swapped, and a double dissociation across domains.
- The honesty split is not used because its `matching` label appears to mark the dishonest answer and the original code defines no honesty prompts.

8 Reproducing

All commands run from the repository root; the scripts need `PYTHONPATH=.`. Calibration and evaluation text:

```
python scripts/make_mmlu_calib.py --out-dir outputs/calib2 \
    math=college_mathematics,... law=professional_law,...
python scripts/make_mmlu_calib.py --split test --out-dir outputs/eval \
    math=... law=...
python scripts/make_persona_calib.py Persona/data/wealth_train.csv \
    --out-dir outputs/persona_calib
```

Statistics, errors, plans and mean ablation:

```
python -m compress collect_stats --model-path M \
    --calib math=... --calib law=... --out stats.pt
python -m compress estimate_quant_error --model-path M --stats stats.pt \
    --row-group 64 --out errors.json
python -m compress allocate_bits --errors errors.json --target math \
    --signal tarc --lam 10 --size-budget N --out plan.json
python -m compress causal_probe --model-path M --stats stats.pt \
    --target math_a --others law_a --k-frac 0.01 \
    --eval-target math=outputs/eval/math.txt \
    --eval-other law=outputs/eval/law.txt --out probe.json
```

Persona experiments (`T` is `Persona/data/wealth_test_infer.csv`):

```
python scripts/persona_gate.py --model-path Qwen/Qwen3.5-0.8B \
    --test T --out gate.json
python scripts/persona_patch.py --model-path Qwen/Qwen3.5-0.8B \
    --stats stats_persona.pt --test T --k-frac 0.05 --per-question \
    --out patch.json                 # add --reverse for necessity
python scripts/persona_quant.py --model-path Qwen/Qwen3.5-0.8B \
    --stats stats_persona.pt --test T --out quant.json
```

A Code map

file	role
`compress/stats.py`	per-domain moments A, V, M
`compress/quant.py`	formats, exact bytes, D, importance, row-group units
`compress/allocate.py`	estimation, greedy plan, contrast signals (including TARC)
`compress/serialize.py`	packed storage, manifest, grouped layers
`compress/patching.py`	channel sets, ablation hooks, dose-matched controls
`compress/pair_patching.py`	minimal-pair patching, restoration, necessity
`compress/quant_probe.py`	column-wise fake quantization and retention
`scripts/persona_*.py`	persona gate, patching and quantization drivers

References

[1] M. Sun, Z. Liu, A. Bair, J. Z. Kolter. A simple and effective pruning approach for large language models. 2023.

[2] T. Dettmers, A. Pagnoni, A. Holtzman, L. Zettlemoyer. QLoRA: efficient finetuning of quantized LLMs (NF4). 2023.

[3] K. Meng, D. Bau, A. Andonian, Y. Belinkov. Locating and editing factual associations in GPT. 2022.

[4] "Your Language Model Secretly Contains Personality Subnetworks", the paper whose code is kept in `Persona/`.